EMIRATES GROUP HR GOVERNANCE

OFFICIAL HACKATHON SUBMISSION
Ignyte x Human-in-the-Loop • Continuous HR Compliance Challenge

Continuous HR Compliance Intelligence Platform

SOLUTION NOTE • 2-PAGE EXECUTIVE BRIEF & RESPONSIBLE-AI STATEMENT

100%	18	< 2.5s	AED 14.8M
COVERAGE (ZERO SAMPLING)	CODIFIED EMIRATES RULES	10,000 TXN BATCH LATENCY	FINANCIAL EXPOSURE GUARDED

1. Problem Statement: Moving Beyond the Manual Sampling Trap

Emirates Group HR processes tens of thousands of complex monthly transactions across Emirates Airline, dnata ground operations, cargo, flight catering, and corporate services: hires, allowances, flight crew certifications, lateral transfers, and exits. Currently, compliance relies on periodic manual sampling (5% to 10%). Consequently, **breaches surface late** (often weeks after payroll lock), **audit coverage is uneven** with systemic blind spots in Delegation of Authority (DOA) and Segregation of Duties (SoD), and **audit readiness requires frantic, manual effort**. As transaction volumes surge, this manual approach does not scale.

2. What We Built: Dual-Engine Continuous Compliance Architecture

We built a production-ready system evaluating **100% of transactions in real time** before payroll execution. The system features a dual-track analytical architecture:

Track A: Deterministic Rule Engine (18 Codified Policies)
Evaluates every transaction against mathematically formal policy thresholds:

- **Delegation of Authority (`POL-DOA-001`):** Grade-based financial ceilings (Grade ≤08: AED 10k, Grade 09–10: AED 25k, Grade 11–12: AED 60k, Grade 13+: Executive Committee approval).
- **Segregation of Duties (`POL-SOD-001`, `POL-SOD-002`):** Zero-tolerance prohibition of self-approval and subordinate approving senior supervisor.
- **Mandatory Statutory & Aviation Clearance (`POL-DOC-001–004`):** DHA medical certificates for sick leave >2 days, attested certificates for dependents, exit clearance handover, and **GCAA Class 1 Medical Fitness** for flight deck crew.
- **Timing & Retrospective Windows (`POL-TIM-001–003`):** 30-day retroactive effective date ceiling (Group HR SVP waiver required), 12-month tenure in-role prior to lateral transfer, and 30/90-day separation notice compliance.
- **Remuneration Integrity (`POL-ALLOW-001/002`, `POL-SAL-001`, `POL-DUP-001`):** Grade housing allowance caps, AED 45,000 child education ceiling, 15% off-cycle salary hike cap, and duplicate allowance detection across 30-day rolling cycles.

Track B: Unsupervised Machine Learning Anomaly Detector (Isolation Forest & Z-Scores)
Flags behavioral exploits beyond written rules across 9 feature dimensions: approver velocity bursts (≥12 approvals in 7 days indicating rubber-stamping), beneficiary surges (4+ requests in 30 days to bypass ceilings), departmental financial outliers (>3.0 standard deviations), and off-hour weekend filings.

3. Explainable AI, Evidence Matrix & Intelligent Routing

Instead of cryptic system error codes, our explainability layer generates: (1) **Conversational Explanation** in plain English and Arabic (العربية) with full RTL support; (2) **Cited Evidence Matrix** providing exact mathematical comparisons (e.g., `amount=AED 18,500; approver_grade=8; max_threshold=AED 10,000; excess=AED 8,500`); (3) **Severity Score** (CRITICAL, HIGH, MEDIUM, LOW); (4) **Direct Operational Dispatch** to domain owners (Compensation & Benefits Lead, Payroll Operations, Talent Mobility, Aviation Regulatory HR, or Internal Audit); (5) **Prescribed Corrective Action** guiding analysts to resolve the root cause.

4. Responsible AI, Security & Human-in-the-Loop Statement

> **The Inviolable Guardrail: The Tool Recommends; An Authorized Human Decides**
> The platform **never mutates, cancels, or approves transactions automatically**. It operates strictly as high-speed decision intelligence. Master HR transaction tables are never altered without human authorization.

Audit Trail & Immutability: Every flagged exception logs an immutable ISO timestamp, rule version hash, cited data snapshot, and analyst action (`APPROVED_WITH_WAIVER`, `RETURNED_TO_ORIGINATOR`, `REMEDIATED`, `ESCALATED_TO_AUDIT`) with mandatory rationale notes. All findings export instantaneously into the required `finding_format.csv` structure.
Zero Generative Hallucination: Policy citations and evidence are generated via deterministic templates bound directly to verified transaction fields, eliminating LLM hallucination risks.
Data Privacy & UAE Sovereignty: Evaluated strictly on synthetic data. Production architecture complies with **UAE Federal Decree-Law No. 45 of 2021 on Personal Data Protection (PDPL)**, supporting Role-Based Access Control (RBAC), TLS 1.3 encryption, and PII masking.

5. How the Solution Runs Inside Emirates Enterprise Environment

Cloud Hosting & Data Residency
Deployed inside **Microsoft Azure UAE North (Dubai)** or AWS Middle East (UAE) sovereign region within an air-gapped Emirates Private VNet. Zero data egresses the United Arab Emirates.

HCM Ingestion Integration
Ingests daily delta extracts (SFTP with Customer-Managed Keys) or real-time event webhooks from **Oracle Cloud HCM**, **SAP SuccessFactors**, and **ServiceNow HRSD**.

6. Nice-to-Have Capabilities: NLP Studio, Fairness & Bilingual Interface

No-Code Natural Language Rule Studio: Compliance analysts type new rules in plain English (e.g., *"Flag housing allowance exceeding 12000 AED for Grade 6 or below"*). The NLP engine compiles the text into an executable AST lambda filter, simulates the rule across 10,000 transactions in <300ms (showing 17 matches), and activates it with a single click.
Procedural Fairness & Employee Experience: Replaces adversarial notices with empathetic "Fair Notices" in English and Arabic. Provides a 1-click employee dispute channel with a 5-day grace period and 3-day SLA resolution before payroll deductions occur. Adverse Impact Ratio (AIR) monitoring confirms high scrutiny on executive leadership (93% DOA scrutiny) without penalizing frontline workers (14.2% rate).

7. 90-Day Pilot Readiness Roadmap (Next Quarter Rollout)

Timeline	Phase Focus	Operational Milestones
Weeks 1–3	Ingestion & Shadow Run	Connect read-only Oracle HCM SFTP batch feeds; run shadow audit parallel to manual sampling.
Weeks 4–7	Lead Triage & Tuning	Onboard 8 HR Analysts & 4 Ops Leads; tune departmental thresholds; activate bilingual UI.
Weeks 8–10	Employee Fair Notice	Enable Fair Notice notifications and 1-click appeal channel with 3-day SLA resolution.
Weeks 11–12	Full Production Cutover	Decommission manual sampling; achieve 100% continuous audit coverage across Emirates & dnata.

8. Submission Benchmark Summary

2,761	618	0
EXCEPTIONS FLAGGED (10,000 TXNS)	EXCEPTIONS ON UNLABELLED TEST SET	NULL FIELDS IN FINDING OUTPUT